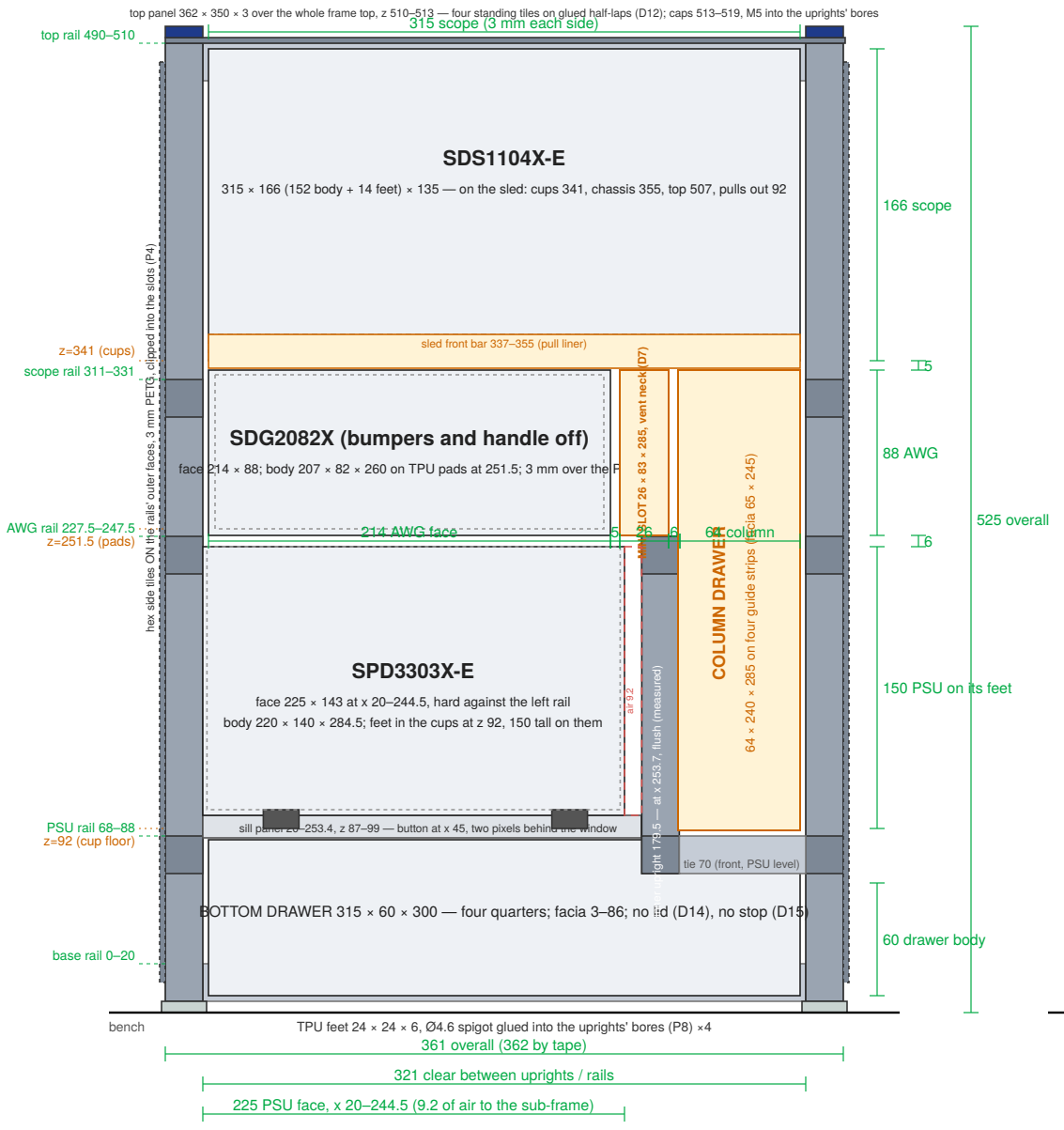
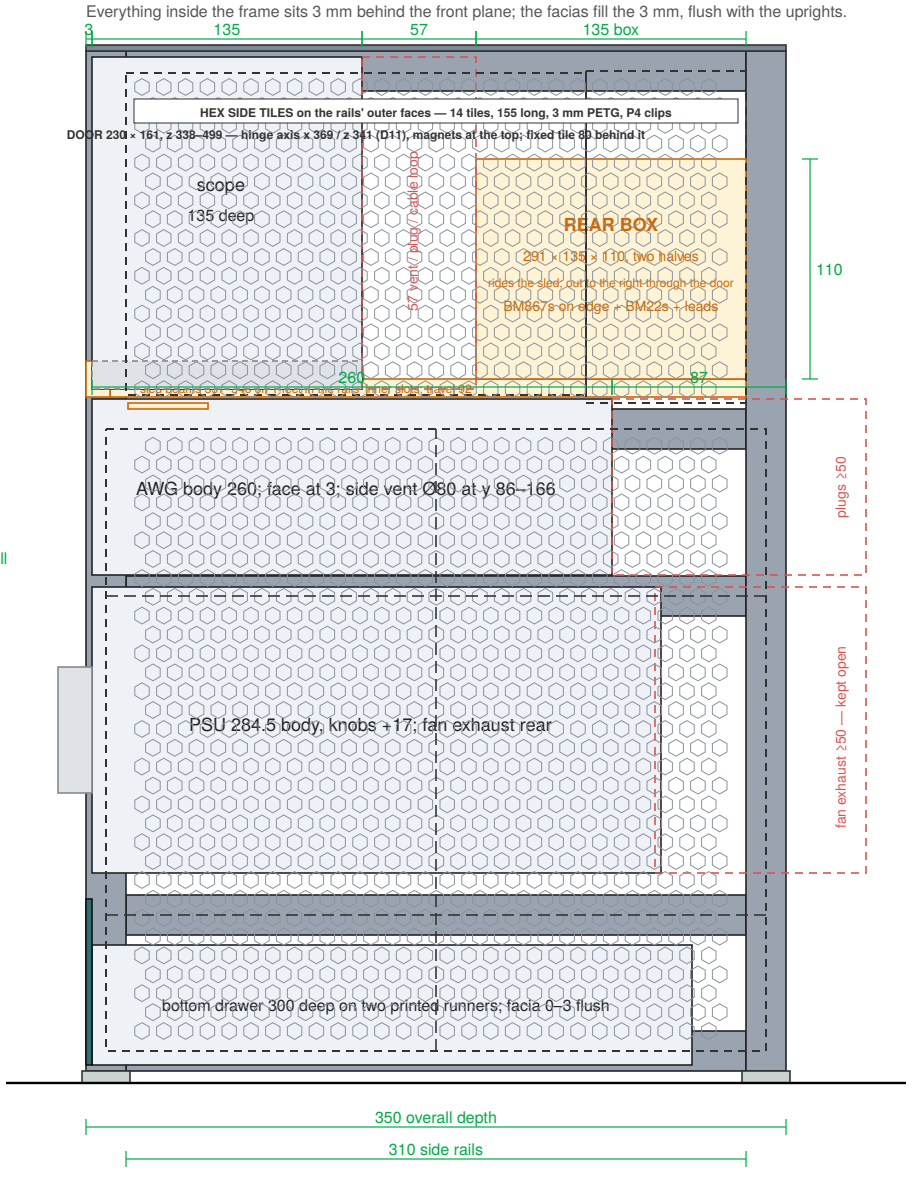
FRONT ELEVATION — as built

1 unit = 1 mm. Frame 361 in the model, 362 by tape. Instruments as measured on the bench; printed parts in orange.



SIDE ELEVATION — right side, front at left

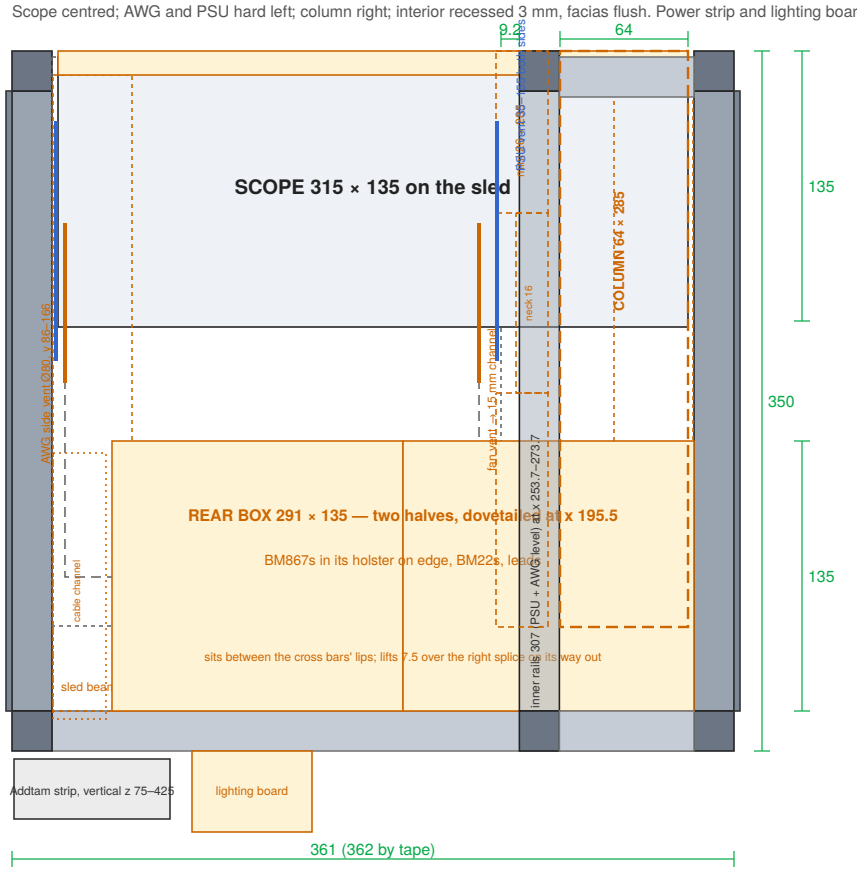
Everything inside the frame sits 3 mm behind the front plane; the facias fill the 3 mm, flush with the uprights.



Rail gaps as built, face to face — right 45.4 / 139.5 / 65.8 / 158, left 51.2 / 135 / 65.5 / 157
(spacers 48 / 139.5 / 63.5 / 159). The model keeps the spacer heights; the side tiles were cut to the measured bands, per side.

PLAN — front at top

Scope centred; AWG and PSU hard left; column right; interior recessed 3 mm, facias flush. Power strip and lighting board behind the frame.



2020 EXTRUSION — AS CUT (CUT-TABLE.md)

Uprights	4	510
Side rails, 5 levels	10	310
Rear cross rails (base + top)	2	321
Inner rails (PSU + AWG level; 2 used, 2 spare)	4	307
Inner uprights	2	179.5
Ties, inner → right rails	3	70
From		8 × 1 m + the CNCMANS 300 mm 4-pack (D1)

Joints: 40 hidden L-connectors on the inward faces + 8 cast brackets (4 under the scope rails, 1 at the front tie);
drop-in T-nuts; M5 × 8 for the brackets, M5 × 10 for the printed parts.

LEVELS AS BUILT (z from the rail bottoms, mm)

Base rails	0–20	feet −6–0; bottom drawer body 3–63 on runners in the base rails' top slots
PSU rails	68–88	cradle cups floor at 92; PSU on its feet 92–242; sill panel 87–99 under the PSU
AWG rails	227.5–247.5	5 mm cradle arms + 4 mm TPU pads → AWG body 251.5–333.5, face 248–336
Scope rails	311–331	T-feet in the inner slots; beams 337–346, front bar to 355; cups 341, chassis 355, top 507; box 346–456
Top rails	490–510	top panel 510–513 over the whole frame; caps 513–519
Inner sub-frame	68–247.5	x 253.7–273.7 (measured), front upright flush; column 274–338, mini slot 242–268

PRINTED PARTS AS FITTED — 76 designs, 103 pieces (HANDOVER v19 §10)

- Cradles 8: PSU left v3 ×2, right rear v4, right front v5; AWG left v2 ×2, right v2 ×2 — black ABS, seated with the jigs
- Drawers: bottom quarters Q1–Q4 + facia with the logo inlay + pull insert v2 + two 300 runners (ABS+); column drawer + facia + insert + four guide strips; mini slot body + runner v3 + facia + insert
- Sled: two 210 T-feet, four beam halves (split, D6), front bar v3, cross bars v2, front rests v2, splices v2, cable channel, pull liner, two TPU stop caps
- Box: two halves (dovetailed) + two corner braces. Skin: 14 side tiles (bands at the measured heights), door v3 + two hinge pairs + pins, fixed rear tile, front bracket, rear tab
- Top: four standing tiles + four caps. Trim: 4 TPU feet, 4 AWG pads, 6 slot covers, 2 strip brackets, sill panel + window + 2 blocks, board enclosure + lid

AS BUILT — WHERE THE BENCH OVERRULED THE MODEL

- Sub-frame upright at x 253.7, flush with the front (model 251, 3 back) — only the sill panel bridges to it, so only the sill panel changed
- Rail gaps up to 6 mm off the spacers → every side tile cut to its own band per side (left 71.2 / 155 / 85.5, right 65.4 / 159.5 / 85.8)
- PSU feet 155.5 apart and 150 tall on them → right cradles v4 / v5, AWG arms 8 → 5 mm; the PSU sits very slightly skewed (accepted)
- Sled front droops past the foot → front rests on the rail tops and splice bars over the P7 joint; the box lifts 7.5 over the right splice
- Door edge 347 → 338 (7 over the scope rail); no drawer lids (D14) and no drawer stops (D15); strips 15 LEDs each, not 28; sled travel 92 to the TPU caps
- Not built, by choice: the probe hanger and the rear-right cable clips